SEC SchedX

Complete Model Guide

From sign-in and college setup to staged timetable generation, review, and PDF export.

Sudharsan Engineering College | Local application guide

Based on the current application source and local database snapshot. This guide describes model behavior and current readiness; it contains no account passwords.

Local app: http://127.0.0.1:3000/login | Backend health: http://127.0.0.1:8000/health
Covered: roles, setup, curriculum, generation, timetable views, PDF designer, exports, current data, and operating checklist.

Use the email and password assigned to your institution account.

1. Product, sign-in, and navigation

SEC SchedX manages departments, faculty, class sections, semester subjects, schedule rules, staged timetable generation, review, and printable output. The frontend uses Next.js 14 / React 18; backend uses FastAPI, SQLAlchemy, and SQLite in this local environment; timetable generation uses OR-Tools CP-SAT.

Sign-in

Step	Action / result
1	Open http://127.0.0.1:3000/login and enter your assigned email/password.
2	Select Sign in. Successful sign-in opens the principal workspace; invalid credentials show an error.
3	Use the role-filtered sidebar. Sign out from the account controls; expired sessions return to sign-in.

Role-based access

Role	Typical visible access
Super administrator	Workspace setup, departments, faculty, subjects, classes, generation, timetable, master view, settings.
HOD	Department operations, curriculum, faculty, class setup, generation, timetable, and settings.
Faculty	Overview, faculty schedules, timetable viewing, settings, and About.
Student	Overview, timetable viewing, settings, and About.

Main sections

Departments and Faculty maintain people and organizational records. Curriculum & Subjects maps what each class studies. Class settings defines sections. Generate builds schedules in order. Timetable is the class weekly view; Master shows multiple classes by day. Settings controls academic rules, appearance, and export defaults. Import handles data spreadsheets separately from template import.

2. Prepare the institution and curriculum

Complete data setup before generating. A class needs subject allocations; each subject allocation needs an assigned teacher. Missing records block or reduce generation.

Settings and section counts

- Review college name/code, affiliation, principal, address, email, website, academic year, language, time format, and theme. Save workspace settings.
- Set default semester cycle. Odd is semesters 1, 3, 5, 7; Even is 2, 4, 6, 8. Class setup supports Both, Odd, or Even; generate each cycle separately.
- In Settings > Class settings, choose department, cycle, and counts for Years 4, 3, 2, and 1; save in Year 4 to Year 1 order. Sections are A, B, C, etc.
- Reducing section counts asks for confirmation and removes selected classes, their subject allocations, and generated timetable entries. Review that list before confirming.

Faculty and subjects

- Create faculty records with correct home department and teaching eligibility. Verify sample/demo names before using them in an official schedule.
- In Curriculum & Subjects select department type, department, semester, and section when multiple classes exist. Add subject name/code, theory or lab type, credits, weekly periods, and faculty.
- For practical/lab subjects, choose Main or Integrated mode and consecutive block size. Weekly lab periods must be divisible by the block size.
- Check semesters 1 through 8 where applicable. Confirm each configured class has subjects and every allocated subject has a faculty member.

Schedule structure defaults and controls

Control	Current / effect
Working days	Monday-Friday currently; controls solver availability and displayed days.
Periods	7 per day, 50 minutes, starts 09:00; current weekly target is 35.
Breaks	Short break after P2 for 15 minutes; lunch after P4 for 45 minutes. Master can show a break after P6.
PDF branding	Official Class Timetable; class/master export defaults Modern; footer Generated by SEC SchedX.

Save all settings before generation. Verify subject workload fits available weekly slots and faculty are assigned.

3. Generate the timetable in stages

Choose department type, department, and cycle on Generate. The screen shows configured sections and which one is ready. Prerequisite order prevents downstream stages from relying on incomplete schedules.

Cycle	Semester order	Year order
Odd	7 -> 5 -> 3 -> 1	Year 4 -> 3 -> 2 -> 1
Even	8 -> 6 -> 4 -> 2	Year 4 -> 3 -> 2 -> 1

Within each year, generate Section A then B, C, etc. Complete every section of that year before starting the next year. Later sections reference earlier sections; lower years reference all completed earlier semesters.

Operator steps

- Select type, department, and Odd/Even cycle; confirm the configured years/sections.
- Generate Year 4 Section A, review it, then continue with all remaining Year 4 sections.
- Proceed through Year 3, Year 2, and Year 1. Repeat as a separate workflow for the other cycle.
- For a paired Lab department, use Generate Lab Timetable; lab generation uses matching literature-department classes and lab subjects.
- Use View for review, Regenerate to rerun, or Delete to clear a section and dependent later schedules.

Solver checks and downstream effects

- Backend checks workflow order, completed references, faculty availability, and timetable conflicts before commit. Missing faculty, over-capacity workloads, or infeasible labs can block generation.
- Regenerating an earlier stage can clear later sections in the same year and lower-year schedules that depend on its faculty reservations. Rebuild affected stages in sequence.

Before regenerating or deleting an upstream stage, plan to rebuild its dependent sections and years in order.

4. View, edit, and export timetable data

Class timetable

Choose department type, department, Odd/Even cycle, and class. The standard view shows the weekly grid. If no timetable exists, the page links to Generate.

Control	Behavior
Show timetable with template	Shows the selected grid structure. Grid cells show subject and faculty; subject details, credits, department, coordinator, and approval tables can appear below.
Breaks in PDF	Adds configured break/lunch columns to a class PDF when checked.
Manual edit	Add, replace, move, or clear a period subject to weekly limits and faculty conflict rules.
Generate or regenerate	Returns to staged generation with selected class context.
Export timetable PDF	Opens print-ready output; choose Save as PDF in the browser print dialog.

Faculty timetable

Review an individual faculty schedule and filter by Odd/Even. Faculty availability is used when assigning and generating. Complete home-department schedules before relying on cross-department teaching availability.

Master timetable

Select department type, department, semester cycle, and weekday. The grid shows class, period, subject code/name, and faculty. Choose Show configured breaks to display breaks and include them in the master PDF. For seven periods, a break appears after Period 6 when another break is not already there.

Master export can print the selected day or all days; all-days output uses one page per weekday. Class and master use separate saved template defaults.

Existing schedules are present for CIVIL Semester 7, CSE Semester 7, and EEE Semesters 5 and 7. Regenerating those stages can affect dependent schedules.

5. PDF template designer and export

The template is a reusable layout. It supplies page styling and grid structure; class/master timetable entries are filled from the selected schedule when exported.

Design and save

- Open Settings > PDF template designer. Choose Modern, Classic, or Minimal; set separate class and master defaults, accent color, footer, page position, and background visibility.
- DOCX import reads a DAY / PERIOD grid and recognized break/lunch columns; following tables may provide subject, coordinator, and approval labels.
- PDF, PNG, and JPG are page backgrounds. The grid/data are rendered separately over the background; these formats do not automatically parse timetable cells.
- Select Save PDF template after editing. CSV/XLSX record import is a separate validate/preview/confirm workflow.

Input	How it is used	Limit
DOCX	Grid structure and break/lunch labels; may read following detail/sign-off tables.	Under 5 MB; DAY / PERIOD grid required.
PDF	Background page behind generated timetable.	Under 1.5 MB.
PNG/JPG	Logo, border, or page background.	Under 1.5 MB.
CSV/XLSX	Bulk records through Import, with validation before confirmation.	Use correct entity template.

Class PDF export

Select a class, optionally review Show timetable with template, check Breaks in PDF if needed, then select Export timetable PDF. In the browser print dialog choose Save as PDF. Review orientation, scale, signatories, and page breaks before saving.

Master PDF export

Select a department/day, choose Show configured breaks if wanted, then export the selected weekday or all weekdays.

Current template state: a background image is saved, class/master defaults are Modern, and no DOCX grid structure is saved. Confirm layout in the live preview before official export.

6. Live data and readiness snapshot

Read-only local database snapshot dated 04 October 2026. The earlier live workflow trial used temporary data that was removed; the existing 165 schedule entries were preserved.

Inventory	Count
Academic departments	5
Lab departments	3
Classes	52
Subjects	370
Faculty	39
Scheduled entries	165
Subject allocations missing faculty	75: AI and DS 74; CSE 1
Classes without subject allocations	9: AI and DS 8; EEE 1

Department status

Department	Cls	Staff	Subj.	Slots	Readiness
AI and DS (AI&DS)	16	6	74	0	74 missing teacher links; 8 empty classes.
AIML	8	8	74	0	Assignments have faculty; no timetable entries.
CIVIL	8	8	74	33	Year 4, Semester 7, Section A scheduled.
CSE	8	9	74	33	Year 4, Semester 7, Section A; one missing teacher link.
EEE	12	8	74	99	Year 4 Semester 7 A/B and Year 3 Semester 5 A; one empty class.

Live generation trial

A temporary department completed Section A for all eight semesters (Odd and Even), then generated A/B sections in the requested order. All 24 small test generation calls succeeded. The temporary classes, subjects, schedules, and test faculty were deleted afterward.

This confirms staged generation on a small clean curriculum, not feasibility of the full 370-subject curriculum or lab workloads. Complete 75 missing faculty links and 9 empty class allocations before a college-wide build.

7. End-to-end operator checklist

Setup readiness

- [ ] Sign in with an administrator/HOD account; confirm institution profile and academic year.
- [ ] Verify departments, lab pairings, faculty names, home departments, and teaching eligibility.
- [ ] Save section counts by year and semester cycle.
- [ ] Assign every subject, credits, weekly periods, lab mode/block size, section scope, and teacher.
- [ ] Set working days, daily periods, start time, duration, break/lunch, weekly target, and PDF defaults.

Generate, review, export

- [ ] Generate Odd Year 4 through Year 1, all sections in order; repeat for Even.
- [ ] Generate linked lab timetable if required; resolve solver errors before downstream stages.
- [ ] Review class views, faculty schedules, and master day views for gaps, overloads, and conflicts.
- [ ] After changing an upstream stage, rebuild dependent sections/years in sequence.
- [ ] Confirm PDF background, grid, break columns, detail/signatory tables, and page settings.
- [ ] Export one class proof and one master proof. Choose Print > Save as PDF; inspect before final distribution.

Quick recovery

Issue	Action
No class sections	Save Class settings for this department/year/cycle and refresh Generate.
Missing subject/faculty	Complete class-subject assignments and teacher links.
Order/prerequisite error	Finish earlier year/section before continuing.
Faculty conflict or infeasible workload	Check shared schedules, weekly periods, and slot capacity.
Template shows background only	Import DOCX for grid parsing; PDF/images serve as backgrounds.
PDF does not open	Allow popups and use the browser print dialog.

Model stack: Next.js 14 / React 18; FastAPI, SQLAlchemy, and SQLite; OR-Tools CP-SAT. Workspace settings persist through the backend; PDF pages are assembled in the browser and passed to its print dialog.